# Boss Challenge — Paper 34 (Class 7)

Weather, Climate & Adaptations | Physical & Chemical Changes | Arithmetic Expressions | Rational Numbers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

**1.** The day-to-day state of the atmosphere at a place - its heat, rain and wind - is called the:

**(A)** climate
**(B)** weather
**(C)** season
**(D)** habitat

**2.** When something changes but not a single new substance is formed, the change is called a:

**(A)** physical change
**(B)** permanent change
**(C)** nuclear change
**(D)** chemical change

**3.** In the arithmetic expression 7 + 4 + 9, the numbers 7, 4 and 9 are each called a:

**(A)** factor
**(B)** product
**(C)** sum
**(D)** term

**4.** Any number you can express as p/q, with p and q whole numbers and the bottom q not zero, is a:

**(A)** irrational number
**(B)** prime number
**(C)** natural number
**(D)** rational number

**5.** The average pattern of the weather of a place taken over many years is called its:

**(A)** forecast
**(B)** weather
**(C)** climate
**(D)** temperature

**6.** A change in which a brand-new substance with new properties is formed is called a:

**(A)** reversible change
**(B)** temporary change
**(C)** physical change
**(D)** chemical change

**7.** Following the correct order, the value of the expression 2 + 3 x 4 is:

**(A)** 24
**(B)** 20
**(C)** 14
**(D)** 9

**8.** The number 0 is a rational number because it can be written as:

**(A)** only a negative number
**(B)** a number that cannot be a fraction
**(C)** 1/0 (one over zero)
**(D)** 0/1 (zero over one)

**9.** The element of weather that a thermometer is used to measure is the:

**(A)** rainfall
**(B)** temperature
**(C)** humidity
**(D)** wind speed

**10.** The melting of a block of ice into water is an example of a:

**(A)** burning change
**(B)** physical change
**(C)** irreversible change
**(D)** chemical change

**11.** With the bracket worked out first, the value of the expression (2 + 3) x 4 is:

**(A)** 9
**(B)** 14
**(C)** 24
**(D)** 20

**12.** Written in its simplest (standard) form, the rational number 4/8 is:

**(A)** 8/4
**(B)** 1/2
**(C)** 2/1
**(D)** 4/8 is already simplest

**13.** The chief source of energy that drives all the changes in the weather is the:

**(A)** wind
**(B)** Moon
**(C)** Sun
**(D)** ocean

**14.** The burning of a piece of paper or wood is an example of a:

**(A)** change of state only
**(B)** chemical change
**(C)** physical change
**(D)** reversible change

**15.** In any expression that mixes brackets and other operations, the part you must work out first is:

**(A)** the left-most number, whatever it is
**(B)** the largest number in the expression
**(C)** whatever is inside the brackets
**(D)** the addition, always before brackets

**16.** The additive inverse (opposite) of the rational number 3/7 is:

**(A)** -3/7
**(B)** 0
**(C)** 3/7 itself
**(D)** 7/3

**17.** The amount of water vapour, or moisture, present in the air is known as the:

**(A)** humidity
**(B)** rainfall
**(C)** temperature
**(D)** pressure

**18.** For an iron object to rust, both of these must be present together:

**(A)** strong sunlight and shade
**(B)** air (oxygen) and water (moisture)
**(C)** only oil, with no air
**(D)** only dry air, with no water

**19.** Following the correct order, the value of the expression 10 - 2 x 3 is:

**(A)** 30
**(B)** 4
**(C)** 24
**(D)** 11

**20.** Comparing the two negative numbers -1/2 and -3/4, the GREATER one is:

**(A)** neither can be compared
**(B)** they are equal
**(C)** -1/2
**(D)** -3/4

**21.** The rain gauge is the instrument used to measure the amount of:

**(A)** air temperature
**(B)** wind direction
**(C)** rainfall
**(D)** sunshine

**22.** The reddish-brown flaky substance that forms on damp iron is called:

**(A)** rust
**(B)** curd
**(C)** chalk
**(D)** ash

**23.** Following the correct order, the value of the expression 5 x 2 + 3 is:

**(A)** 25
**(B)** 13
**(C)** 30
**(D)** 10

**24.** Every integer, such as 5, is also a rational number because it can be written as:

**(A)** 5/1 (the integer over one)
**(B)** 0/5 (zero over the integer)
**(C)** a number that is never rational
**(D)** 5/0 (the integer over zero)

**25.** The polar bear is kept warm in its freezing home mainly by its thick fur and an inner layer of:

**(A)** scales
**(B)** feathers
**(C)** fat (blubber)
**(D)** sweat glands

**26.** Coating an iron object with a layer of zinc to stop it rusting is called:

**(A)** condensation
**(B)** crystallisation
**(C)** galvanisation
**(D)** evaporation

**27.** Adding zero to a number, as in the expression 8 + 0, always gives:

**(A)** the same number, 8
**(B)** 16
**(C)** 0
**(D)** 80

**28.** Turned upside down, the reciprocal (multiplicative inverse) of 2/3 comes out as:

**(A)** 0
**(B)** 3/2
**(C)** -2/3
**(D)** 2/3 itself

**29.** A polar bear's white fur is useful in the snowy polar region because it helps the bear to:

**(A)** stay hidden (camouflage) against the snow
**(B)** attract more sunlight for warmth
**(C)** look bigger to other bears
**(D)** frighten away the Sun

**30.** Painting or oiling an iron gate helps prevent rust because the layer:

**(A)** keeps air and moisture away from the iron
**(B)** turns the iron into zinc
**(C)** makes the iron heavier
**(D)** cools the iron down

**31.** Following the correct order, the value of the expression 6 + (4 - 1) is:

**(A)** 3
**(B)** 11
**(C)** 9
**(D)** 1

**32.** Adding a rational number to its additive inverse, such as 4/9 + (-4/9), always gives:

**(A)** 4/9
**(B)** 8/9
**(C)** 1
**(D)** 0

**33.** To survive the bitter cold, penguins are often seen to:

**(A)** sleep through the whole winter
**(B)** spread far apart from one another
**(C)** dig deep burrows under the ice
**(D)** huddle closely together in large groups

**34.** The process used to obtain large, pure crystals of a substance from its solution is called:

**(A)** rusting
**(B)** filtration
**(C)** galvanisation
**(D)** crystallisation

**35.** With the bracket worked out first, the value of the expression 6 - (4 - 1) is:

**(A)** 3
**(B)** 9
**(C)** 1
**(D)** 11

**36.** Reduced to its simplest form, the rational number -8/12 is:

**(A)** 2/3
**(B)** -4/6
**(C)** -2/3
**(D)** -12/8

**37.** The climate of the polar regions, such as the Arctic, can best be described as:

**(A)** mild with four equal seasons
**(B)** hot and dry like a desert
**(C)** very cold and ice-covered for most of the year
**(D)** warm and rainy all year

**38.** Dissolving a spoon of sugar in a glass of water is an example of a:

**(A)** physical change
**(B)** irreversible change
**(C)** burning change
**(D)** chemical change

**39.** Using the distributive rule, the value of the expression 4 x (10 + 2) is:

**(A)** 48
**(B)** 16
**(C)** 42
**(D)** 24

**40.** On a number line, the rational number -2/3 is found:

- **(A)** between 0 and -1, on the left of zero
- **(B)** between -1 and -2
- **(C)** exactly on the number 1
- **(D)** between 0 and 1, on the right of zero

**41.** The climate of a tropical rainforest, such as near the equator, is best described as:

- **(A)** hot and wet (humid) for most of the year
- **(B)** hot and very dry
- **(C)** cold and snowy
- **(D)** cool and windy

**42.** The cooking of raw food, such as baking a cake from batter, is an example of a:

- **(A)** change of state only
- **(B)** physical change
- **(C)** chemical change
- **(D)** reversible change

**43.** Removing the bracket, the expression 20 - (5 + 3) has the value:

- **(A)** 28
- **(B)** 12
- **(C)** 22
- **(D)** 18

**44.** A rational number that lies exactly halfway between 0 and 1 is:

- **(A)** 2 (two)
- **(B)** 1/4
- **(C)** 0 itself
- **(D)** 1/2

**45.** A special feature of a living thing that helps it survive in its surroundings is called an:

- **(A)** adaptation
- **(B)** imitation
- **(C)** invention
- **(D)** decoration

**46.** The souring of milk to form curd (dahi) is an example of a:

- **(A)** reversible change
- **(B)** physical change
- **(C)** change of state only
- **(D)** chemical change

**47.** Comparing the two expressions 3 + 4 and 3 x 4, the one with the GREATER value is:

- **(A)** they are exactly equal
- **(B)** 3 + 4, which equals 12
- **(C)** 3 + 4, which equals 7 and is bigger
- **(D)** 3 x 4, which equals 12

**48.** Multiplying a rational number by its reciprocal, such as 5/6 x 6/5, always gives:

- **(A)** 11/11 written as a fraction
- **(B)** 1
- **(C)** 0
- **(D)** 25/36

**49.** Some birds fly thousands of kilometres to warmer places to escape the harsh cold of winter. This is called:

- **(A)** migration
- **(B)** germination
- **(C)** hibernation
- **(D)** camouflage

**50.** When a strip of magnesium ribbon is burnt in air, it gives a white powder. This shows a:

- **(A)** change of state only
- **(B)** reversible change
- **(C)** physical change
- **(D)** chemical change

**51.** Following the correct order, the value of the expression 12 / 3 + 1 is:

- **(A)** 13
- **(B)** 5
- **(C)** 4
- **(D)** 3

**52.** Comparing the two rational numbers 3/4 and 2/3, the GREATER one is:

- **(A)** they are equal
- **(B)** 2/3
- **(C)** 3/4 only when both are negative
- **(D)** 3/4

**53.** The highest temperature reached during a day is called the day's:

- **(A)** maximum temperature
- **(B)** average temperature
- **(C)** minimum temperature
- **(D)** normal temperature

**54.** The white powder formed when magnesium burns is dissolved in water. The solution turns out to be:

- **(A)** basic (turns red litmus blue)
- **(B)** neutral like pure water
- **(C)** acidic (turns blue litmus red)
- **(D)** salty like sea water

**55.** Following the correct order, the value of the expression 2 x (5 + 5) is:

- **(A)** 17
- **(B)** 15
- **(C)** 12
- **(D)** 20

**56.** Written with a denominator of 6, the rational number equal to 1/2 is:

- **(A)** 6/3
- **(B)** 2/6
- **(C)** 1/6
- **(D)** 3/6

**57.** The lowest temperature reached during a day, usually in the early morning, is the day's:

- **(A)** minimum temperature
- **(B)** boiling temperature
- **(C)** average temperature
- **(D)** maximum temperature

**58.** Tearing a sheet of paper into small pieces is an example of a:

- **(A)** chemical change
- **(B)** burning change
- **(C)** irreversible chemical change
- **(D)** physical change

**59.** Following the correct order, the value of the expression 100 - 10 x 5 is:

- **(A)** 550
- **(B)** 450
- **(C)** 85
- **(D)** 50

**60.** The temperature one morning is -2 °C and it rises by 5 °C by noon. Written as a sum of signed numbers, -2 + 5 gives:

- **(A)** 7 °C
- **(B)** -7 °C
- **(C)** 3 °C
- **(D)** -3 °C

**61.** The polar bear's broad, large paws are useful because they help it to:

- **(A)** dig deep tunnels for shelter
- **(B)** fly short distances
- **(C)** walk on snow and swim without sinking
- **(D)** climb tall forest trees

**62.** Most physical changes are easy to REVERSE, as shown by the way that:

- **(A)** rust can be turned back into shiny iron just by drying
- **(B)** burnt wood can be turned back into fresh wood
- **(C)** water can be frozen to ice and the ice melted back to water
- **(D)** curd can be turned back into fresh milk

**63.** Following the correct order, the value of the expression 3 x 3 + 3 is:

- **(A)** 9
- **(B)** 27
- **(C)** 18
- **(D)** 12

**64.** The number -7 is a rational number, and as a fraction it can be written as:

- **(A)** 7/1
- **(B)** -7/0
- **(C)** -1/7
- **(D)** -7/1

**65.** A penguin's smooth, streamlined body shape is an adaptation that mainly helps it to:

- **(A)** move easily and fast through water
- **(B)** dig in the ice
- **(C)** fly high above the sea
- **(D)** stay warm on land

**66.** When carbon dioxide gas is passed through clear lime water, the lime water turns:

- **(A)** milky (cloudy white)
- **(B)** bright red
- **(C)** deep blue
- **(D)** completely colourless and clear

**67.** Following the correct order, the value of the expression 2 + 2 x 2 + 2 is:

- **(A)** 8
- **(B)** 12
- **(C)** 16
- **(D)** 10

**68.** Working it out, the value of the subtraction 1/2 - 1/4 is:

- **(A)** 0
- **(B)** 1/4
- **(C)** 1/2
- **(D)** 2/6

**69.** Compared with the weather, the climate of a place is something that:

- **(A)** changes from hour to hour
- **(B)** changes every single minute
- **(C)** stays much the same over many years
- **(D)** never existed before today

**70.** A useful clue that a CHEMICAL change has taken place is when, during the change, there is:

**(A)** only the object being moved to a new place
**(B)** the giving off of a gas, heat, light or a new colour
**(C)** only the object getting wet
**(D)** only a change in the shape of the object

**71.** The phrase 'seven more than twice six' is written as the expression 2 x 6 + 7, whose value is:

**(A)** 13
**(B)** 84
**(C)** 19
**(D)** 26

**72.** Working it out, the value of the addition (-1/5) + (1/5) is:

**(A)** 0
**(B)** 1/5
**(C)** 2/5
**(D)** -2/5

**73.** On a humid day our sweat dries slowly and we feel sticky because the air already holds:

**(A)** too much dust
**(B)** no oxygen at all
**(C)** very little moisture
**(D)** a lot of water vapour

**74.** Inside green leaves, plants combine carbon dioxide and water in sunlight to make food. This process is a:

**(A)** change of state only
**(B)** physical change
**(C)** chemical change
**(D)** reversible change like melting

**75.** Following the correct order, the value of the expression (8 - 3) x 2 is:

**(A)** 16
**(B)** 10
**(C)** 2
**(D)** 13

**76.** Both -3/5 and 3/5 have the same size but opposite signs, so they are best described as:

**(A)** reciprocals of each other
**(B)** additive inverses of each other
**(C)** exactly equal numbers
**(D)** both positive numbers

**77.** Large herds of elephants, big cats and many colourful birds are found together mainly in the:

- **(A)** dry sandy desert
- **(B)** tropical rainforest
- **(C)** frozen polar region
- **(D)** high snowy mountain top

**78.** Rusting is considered harmful mainly because, over time, it:

- **(A)** turns iron into pure gold
- **(B)** weakens iron objects and slowly eats them away
- **(C)** makes iron shinier and stronger
- **(D)** cools the iron so it lasts longer

**79.** Following the correct order, the value of the expression 9 + 9 / 9 is:

- **(A)** 3
- **(B)** 2
- **(C)** 18
- **(D)** 10

**80.** Working it out, the value of the addition 2/7 + 3/7 is:

- **(A)** 5/14
- **(B)** 1/7
- **(C)** 5/7
- **(D)** 6/7

**81.** A maximum-minimum thermometer is specially useful because in one reading it shows the:

- **(A)** air pressure changes
- **(B)** rainfall and wind speed
- **(C)** highest and lowest temperatures of the day
- **(D)** humidity of the air

**82.** Stainless steel cutlery does not rust easily because it is a special mixture of metals, that is, an:

- **(A)** alloy made to resist rusting
- **(B)** alloy that rusts faster than iron
- **(C)** type of plastic, not a metal
- **(D)** pure single metal like plain iron

**83.** A plant is given (3 + 2) hours of sunlight each day for 4 days. The expression 4 x (3 + 2) gives a total of:

- **(A)** 14 hours
- **(B)** 9 hours
- **(C)** 20 hours
- **(D)** 24 hours

**84.** A rational number is called POSITIVE when its numerator and denominator have:

- **(A)** opposite signs to each other
- **(B)** the same sign (both positive or both negative)
- **(C)** a denominator of zero
- **(D)** a numerator of zero

**85.** A thermometer in the morning reads 24 °C and by afternoon it reads 36 °C. The day's temperature RANGE is:

- **(A)** 24 °C
- **(B)** 60 °C
- **(C)** 12 °C
- **(D)** 36 °C

**86.** The boiling of water in a kettle, turning it into steam, is an example of a:

- **(A)** chemical change
- **(B)** physical change
- **(C)** irreversible change
- **(D)** burning change

**87.** Following the correct order, the value of the expression 3 x 4 - 2 is:

- **(A)** 18
- **(B)** 6
- **(C)** 10
- **(D)** 12

**88.** The product of the two rational numbers 2/3 x 3/2 is:

- **(A)** 5/5 written as a sum
- **(B)** 4/9
- **(C)** 0
- **(D)** 1

**89.** Deep in a tropical rainforest the ground stays dim because the tall treetops form a thick:

- **(A)** sheet of snow on the branches
- **(B)** cloud of sand in the air
- **(C)** canopy that blocks most sunlight
- **(D)** layer of ice over the soil

**90.** Iron nails were kept in three test-tubes: dry air only, boiled water with no air, and ordinary damp air. The nails that rust are those in:

- **(A)** all three test-tubes equally
- **(B)** ordinary damp air, where both air and moisture are present
- **(C)** the boiled water with no air
- **(D)** the dry air only, with no moisture

**91.** Two expressions have the SAME value, so we may join them with an equals sign. This is true for:

- **(A)** 3 + 3 and 2 + 2, which are equal
- **(B)** 2 + 3 and 1 + 4, since both equal 5
- **(C)** 2 + 3 and 1 + 5, which are equal
- **(D)** 4 + 4 and 2 + 3, which are equal

**92.** The rational number that is the additive inverse of 0 is:

- **(A)** 0 itself
- **(B)** 1
- **(C)** -1
- **(D)** there is no such number

**93.** An elephant's large flapping ears are an adaptation that mainly help it to:

- **(A)** hear sounds from underground
- **(B)** frighten away the rain
- **(C)** lose extra body heat and stay cool
- **(D)** store water for the dry season

**94.** A burnt matchstick cannot be made to look new and unburnt again. This tells us that burning is:

- **(A)** a reversible physical change
- **(B)** a change with no new substance
- **(C)** the same as melting wax
- **(D)** an irreversible (chemical) change

**95.** To make the two expressions 6 + __ and 10 equal, the number that goes in the blank is:

- **(A)** 6
- **(B)** 16
- **(C)** 10
- **(D)** 4

**96.** A rational number that lies between the two numbers 1/4 and 3/4 is:

- **(A)** 1/8
- **(B)** 1/2
- **(C)** 1 (one)
- **(D)** 0

**97.** The simple instrument that swings round on a roof to show which way the wind is blowing is a:

- **(A)** thermometer
- **(B)** barometer
- **(C)** rain gauge
- **(D)** wind vane

**98.** As a candle burns, the wax near the flame melts and then the wax vapour burns. This shows that a candle burning involves:

- **(A)** no change of any kind at all
- **(B)** both a physical change (melting) and a chemical change (burning)
- **(C)** only a physical change, nothing more
- **(D)** only a chemical change, nothing more

**99.** Following the correct order, the value of the expression 18 / (3 + 3) is:

- **(A)** 3
- **(B)** 9
- **(C)** 12
- **(D)** 6

**100.** Reduced to its simplest form, the rational number 10/5 equals the whole number:

- **(A)** 10
- **(B)** 2
- **(C)** 5
- **(D)** 1/2